Exp 3a: Study of Cisco Packet Tracer – Installation and User Interface Overview

AIM :

To study the installation and the user interface of the Cisco Packet Tracer simulation tool, and to understand its environment for designing a simple network.

ALGORITHM :

1. Start the experiment.
2. Download Packet Tracer from https://www.netacad.com by logging in with the Cisco Networking Academy credentials.
3. Install Packet Tracer according to the operating system (Windows or Linux).
4. Launch Packet Tracer and identify the components of its user interface (menu bar, main toolbar, workspace tabs, workspace, common tools bar, simulation tabs, network component box and user-created packet box).
5. Study the device-type selection box and the device-specific selection box in the network component box.
6. Observe the Logical and Physical workspaces and the Real-time and Simulation modes.
7. Stop the experiment.

PROCEDURE :

A simulator simulates network devices and their environment. Packet Tracer is a network design, simulation and modelling tool. It allows complex systems to be modelled without dedicated equipment and helps to practise network configuration and troubleshooting on a computer or on an Android / iOS based mobile device. It is available for both Linux and Windows, and its protocols are coded to behave in the same way as they would on real hardware.

Installing Packet Tracer: Go to https://www.netacad.com, log in with the Cisco Networking Academy credentials, click the Packet Tracer graphic and download the package for the operating system.

Windows: The setup comes as a single file (Packettracer_Setup6.0.1.exe). Open the file, accept the license agreement, choose a location and start the installation.

Linux: Ubuntu / Debian users download the Ubuntu file and Fedora / Redhat / CentOS users download the Fedora file. Give the file executable permission with chmod and run it to begin the installation.

```
chmod +x PacketTracer601_i386_installer-rpm.bin
./PacketTracer601_i386_installer-rpm.bin
```

User interface overview: The layout of Packet Tracer is divided into the following components (numbered as in Fig 1):

1. **Menu bar** – the common menu used to open, save, print, change preferences, etc.
2. **Main toolbar** – shortcut icons for commonly used menu options such as open, save, zoom, undo and redo; on the right-hand side is an icon for entering network

information of the current network.

3. **Logical / Physical workspace tabs** – used to toggle between the Logical and Physical work areas.
4. **Workspace** – the area where topologies are created and simulations are displayed.
5. **Common tools bar** – controls for manipulating topologies: select, move layout, place note, delete, inspect, resize shape and add simple / complex PDU.
6. **Real-time / Simulation tabs** – used to toggle between real-time and simulation modes; buttons are also provided to control the time and to capture the packets.
7. **Network component box** – contains all the network and end devices available in Packet Tracer. It has two areas: (a) the device-type selection box, which holds the device categories, and (b) the device-specific selection box, which shows the different device models of the selected category.
8. **User-created packet box** – users can create highly customised packets to test their topology; the results are displayed as a list.

OUTPUT :

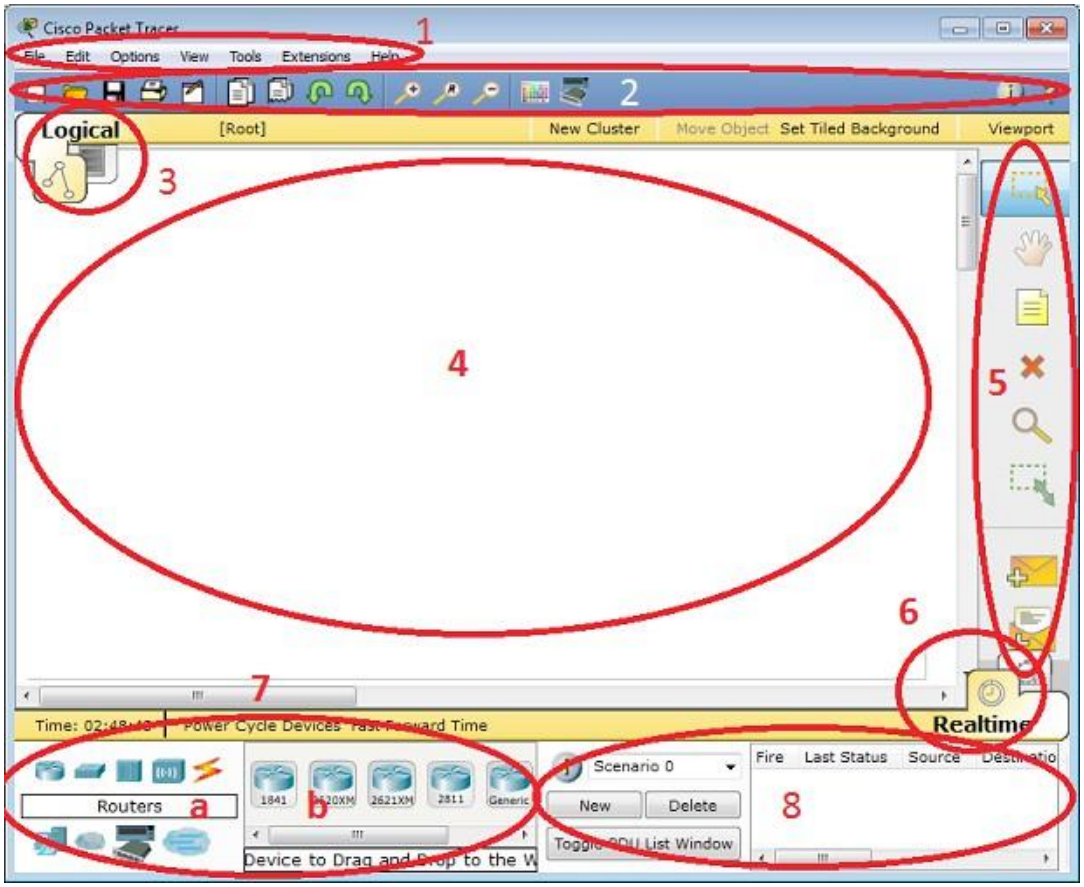


Fig 1. Cisco Packet Tracer – user interface components (1 to 8)

RESULT :

Thus, the installation and the user interface of the Cisco Packet Tracer simulation tool were studied, and its environment for designing a simple network was understood.